

## 26.5 What to do about collinearity

Refitting the same data in another way cannot supply information the design lacks, and every genuine remedy does one of four things: it gets more information of the right kind, changes the question to one the data can answer, adds information from outside the data, or accepts bias in exchange for variance. Before choosing among them, decide whether anything needs to be done at all.

### 26.5.1 Does it matter?

By Theorem 26.1.1(c) and Proposition 26.4.1, collinearity is harmless for quantities with no component along the weak directions: fitted values, predictions that follow the pattern of the data, and combinations such as the sum in Example (Two regressors that move together). If those are the targets, report them and stop.

### 26.5.2 Collect better data

The only remedy that keeps the question is more information along the weak directions. The next proposition shows how much a single well-placed observation can supply.

#### Proposition 26.5.1 — One new observation

Let  $\mathbf{A} = \mathbf{X}^\top \mathbf{X}$  be positive definite with eigenvalues  $\lambda_1 \geq \dots \geq \lambda_p$  and orthonormal eigenvectors  $\mathbf{v}_1, \dots, \mathbf{v}_p$ ,  $p \geq 2$ , and let a case with regressor vector  $\mathbf{x}$  be added, so that  $\mathbf{A}$  becomes  $\mathbf{A} + \mathbf{x}\mathbf{x}^\top$ . Write  $\mathbf{G} = \mathbf{A}^{-1}$ .

- (a) For every  $\mathbf{a}$ , the variance of  $\mathbf{a}^\top \hat{\boldsymbol{\beta}}$  falls by  $\sigma^2(\mathbf{a}^\top \mathbf{G}\mathbf{x})^2/(1 + \mathbf{x}^\top \mathbf{G}\mathbf{x})$ ; in particular it never increases.
- (b) If  $\|\mathbf{x}\| \leq c$ , then  $\lambda_{\min}(\mathbf{A} + \mathbf{x}\mathbf{x}^\top) \leq \min(\lambda_p + c^2, \lambda_{p-1})$ , with equality when  $\mathbf{x} = c\mathbf{v}_p$ .
- (c) Among all  $\mathbf{x}$  with  $\|\mathbf{x}\| = c$ , the choice  $\mathbf{x} = c\mathbf{v}_p$  minimizes the largest variance of  $\mathbf{a}^\top \hat{\boldsymbol{\beta}}$  over unit vectors  $\mathbf{a}$ .

*Proof.*

- (a) By Eq. 1.6.7,  $(\mathbf{A} + \mathbf{x}\mathbf{x}^\top)^{-1} = \mathbf{G} - \mathbf{G}\mathbf{x}\mathbf{x}^\top \mathbf{G}/(1 + \mathbf{x}^\top \mathbf{G}\mathbf{x})$ .
- (b) By Theorem 1.7.7,  $\lambda_{\min}(\mathbf{A} + \mathbf{x}\mathbf{x}^\top) \leq \mathbf{v}_p^\top (\mathbf{A} + \mathbf{x}\mathbf{x}^\top) \mathbf{v}_p = \lambda_p + (\mathbf{x}^\top \mathbf{v}_p)^2 \leq \lambda_p + c^2$ . For the second bound, the plane  $\text{span}\{\mathbf{v}_{p-1}, \mathbf{v}_p\}$  contains a unit vector  $\mathbf{b}$  orthogonal to  $\mathbf{x}$ , since the linear map  $\mathbf{b} \mapsto \mathbf{x}^\top \mathbf{b}$  from the plane to  $\mathbb{R}$  has a nontrivial null space. Then  $\lambda_{\min}(\mathbf{A} + \mathbf{x}\mathbf{x}^\top) \leq \mathbf{b}^\top \mathbf{A} \mathbf{b} \leq \lambda_{p-1}$ . If  $\mathbf{x} = c\mathbf{v}_p$ , then  $\mathbf{A} + \mathbf{x}\mathbf{x}^\top$  has the same eigenvectors as  $\mathbf{A}$  and eigenvalues  $\lambda_1, \dots, \lambda_{p-1}, \lambda_p + c^2$ , whose minimum is  $\min(\lambda_p + c^2, \lambda_{p-1})$ .
- (c) By Theorem 26.1.1(b), the largest variance over unit vectors is  $\sigma^2/\lambda_{\min}$ , and (b) says that  $\mathbf{x} = c\mathbf{v}_p$  maximizes  $\lambda_{\min}$ .

■

The new case should lie along the weakest direction, where the existing data are thin. Such a point breaks the pattern of the regressors, and in observational data it may not exist; in an experiment, or when the sample can be chosen, the proposition says what to look for.

### Example. (The value of one observation)

In the design of Example (Two regressors that move together), add one case at the centred values  $(x_1, x_2) = (1.077, -1.044)$ , the point at distance 1.5 from the centroid along the weak slope direction. Each coordinate is well inside the range of its regressor; only the combination is new. The slope estimates are governed by the slope block with the intercept partialled out (Theorem 6.6.2), and the new case, which has an intercept entry, changes that block from  $\mathbf{S}$  to  $\mathbf{S} + \frac{n}{n+1} \mathbf{x}\mathbf{x}^\top$ , where  $\mathbf{x}$  holds its two slope entries: the columns must be centred again about the new mean, and that costs the factor  $n/(n+1)$ . Applied to this block, Proposition 26.5.1(b) predicts that the smaller eigenvalue rises from 0.172 to  $0.172 + 1.5^2 \cdot 30/31 = 2.349$ , which is what the script finds. The standard deviation of  $\hat{\beta}_1$  falls from  $1.734\sigma$  to  $0.476\sigma$ . To achieve the same reduction with more data of the old pattern, the whole sample would have to be repeated 13.3 times, about 370 further cases in place of one. The script also checks, on the centred block, that no other direction does better.

```
import numpy as np
rng = np.random.default_rng(2601)           # the design of Section 26.1
n = 30
x1 = rng.normal(size=n)
x2 = x1 + 0.15 * rng.normal(size=n)
X = np.column_stack([np.ones(n), x1 - x1.mean(), x2 - x2.mean()])
G = np.linalg.inv(X.T @ X)
lam, V = np.linalg.eigh(X.T @ X)             # ascending eigenvalues
v_weak = V[:, 0]                             # the weak direction (it has
    no intercept part)
```

[Run this code online](#)

```
c = 1.5
x_new = np.r_[1.0, c * v_weak[1:] / np.linalg.norm(v_weak[1:])] # breaks the
    pattern
print("new point (centred x1, x2):", np.round(x_new[1:], 3))
G_new = np.linalg.inv(X.T @ X + np.outer(x_new, x_new))
print("sd of b1 before / after:", round(np.sqrt(G[1, 1]), 3), round(np.sqrt(G_new[1,
    1]), 3))
S_old = X[:, 1:].T @ X[:, 1:]                 # slope block: the intercept
    is orthogonal
xs = x_new[1:]
S_new = S_old + n / (n + 1) * np.outer(xs, xs) # centred slope block after
    recentring
print("smallest slope eigenvalue before / after:", round(np.linalg.eigvalsh(S_old)
    [0], 3), round(np.linalg.eigvalsh(S_new)[0], 3))
# repeating the old pattern k times multiplies X'X by k: Var(b1) falls by the factor
    1/k
k_equiv = G[1, 1] / G_new[1, 1]
print("equivalent number of copies of the whole data set:", round(k_equiv, 1))
```

[Run this code online](#)

### 26.5.3 Change the question

A reparameterization  $\mathbf{X}\boldsymbol{\beta} = (\mathbf{X}\mathbf{T})(\mathbf{T}^{-1}\boldsymbol{\beta})$  leaves the least squares estimate of every linear function of  $\boldsymbol{\beta}$  unchanged (Theorem 6.7.1), so it cannot make  $\beta_1 - \beta_2$  better determined. It can make the reported parameters the well-determined ones. Replacing  $x_1$  and  $x_2$  by their average and their difference gives a coefficient for the average that estimates  $\beta_1 + \beta_2$  precisely, and leaves all the imprecision in the coefficient of the difference. Centring polynomial terms (Example (A quadratic far from the origin)) is the same move.

Two common changes of model are really linear restrictions on  $\boldsymbol{\beta}$ , and should be judged as such. *Dropping a regressor* imposes  $\beta_2 = 0$ . By Theorem 19.1.2 the short slope estimates  $\beta_1 + \pi\beta_2$ , where  $\pi$  is the slope of  $x_2$  on  $x_1$ ; in Example (Two regressors that move together)  $\pi = 1.0261$ , and the short slope has standard deviation  $0.171\sigma$  against  $1.734\sigma$ . It is precise because it answers a different question, giving  $x_1$  the credit for both regressors: with true slopes  $(1, 1)$  it centres at about 2.03. Theorem 19.1.5 gives the condition under which it has the smaller mean squared error. *Combining regressors into an index* imposes a restriction such as  $\beta_1 = \beta_2$ . It concerns the weak direction  $\beta_1 - \beta_2$ , whose estimate has standard deviation  $3.412\sigma$  in the example, so an  $F$  test of it has almost no power: the restriction is cheap in variance and nearly untestable, for the same reason.

### 26.5.4 Add information from outside the data

An exact restriction known from theory is fitted by restricted least squares (Proposition 11.3.3). An approximate one can be written as extra observations  $\mathbf{r} = \mathbf{L}\boldsymbol{\beta} + \mathbf{u}$  and combined with the data by generalized least squares, the mixed estimation of Theil and Goldberger (1961) ([Exercise \(B1\)](#)). The Bayesian version is a prior: with the conjugate prior of Theorem 7.6.2, under which  $\boldsymbol{\beta}$  given  $\sigma^2$  has covariance  $\sigma^2\mathbf{V}_0$ , the posterior mean solves equations with matrix  $\mathbf{V}_0^{-1} + \mathbf{X}^\top\mathbf{X}$ . A prior informative along the weak eigenvector raises the smallest eigenvalue as a new observation does, and the answer along that direction then comes mostly from the prior.

### 26.5.5 Accept bias

The last option gives up unbiasedness. Principal component regression drops the canonical coordinates with small eigenvalues, the truncated SVD estimator of Proposition 10.4.3; ridge regression shrinks every canonical coordinate towards zero, the weak ones most. [Chapter 27](#) develops both (Theorem 27.2.2 and Definition 27.3.1). Neither creates information: they trade variance along the weak directions for bias along the same directions.

### 26.5.6 What not to do

- Do not drop regressors because their variance inflation factors exceed a threshold; the fit may hardly change while the meaning of every remaining coefficient does.
- Do not expect centring or rescaling to cure collinearity that is not created by the parameterization.
- Do not let an automatic search choose among collinear regressors: noise decides which enters ([Chapter 29](#)).

## 26.5.7 Exercises

### 26.5.7.1 A. Check your understanding

#### Exercise A1.

Explain, using Theorem 26.1.1, why no reparameterization of Example (Two regressors that move together) can estimate  $\beta_1 - \beta_2$  with a standard deviation smaller than  $3.412\sigma$ , although the coefficient of the new difference regressor may have any standard deviation you like.

#### Exercise A2.

In Example (Two regressors that move together), what does the short regression of  $y$  on  $x_1$  alone estimate if the true slopes are  $(2, 0)$ ? If they are  $(0, 2)$ ? What does this say about using the short regression to decide which regressor matters?

### 26.5.7.2 B. Practice

#### Exercise B1.

Suppose that, besides  $\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$  with  $\text{Cov}(\boldsymbol{\varepsilon}) = \sigma^2\mathbf{I}$ , there is independent information  $\mathbf{r} = \mathbf{L}\boldsymbol{\beta} + \mathbf{u}$  with  $E(\mathbf{u}) = \mathbf{0}$  and  $\text{Cov}(\mathbf{u}) = \sigma^2\boldsymbol{\Omega}$ ,  $\boldsymbol{\Omega}$  known and positive definite. Show that the generalized least squares estimator from the stacked data is

$$\tilde{\boldsymbol{\beta}} = (\mathbf{X}^\top\mathbf{X} + \mathbf{L}^\top\boldsymbol{\Omega}^{-1}\mathbf{L})^{-1}(\mathbf{X}^\top\mathbf{Y} + \mathbf{L}^\top\boldsymbol{\Omega}^{-1}\mathbf{r}),$$

that it is unbiased with covariance  $\sigma^2(\mathbf{X}^\top\mathbf{X} + \mathbf{L}^\top\boldsymbol{\Omega}^{-1}\mathbf{L})^{-1}$ , and that it equals the posterior mean of Theorem 7.6.2 for a suitable prior. With  $\mathbf{L} = \mathbf{v}_p^\top$ , how does the smallest eigenvalue change?

*Solution.* Stack  $(\mathbf{Y}^\top, \mathbf{r}^\top)^\top = (\mathbf{X}^\top, \mathbf{L}^\top)^\top\boldsymbol{\beta} + (\boldsymbol{\varepsilon}^\top, \mathbf{u}^\top)^\top$ , with error covariance  $\sigma^2 \text{diag}(\mathbf{I}, \boldsymbol{\Omega})$ . Generalized least squares (Corollary 7.2.3) gives the stated estimator, and it is unbiased with covariance  $\sigma^2$  times the inverse of the weighted cross-product matrix. If  $\mathbf{L}$  is square and nonsingular, the prior  $\boldsymbol{\beta} \sim N(\mathbf{L}^{-1}\mathbf{r}, \sigma^2\mathbf{L}^{-1}\boldsymbol{\Omega}(\mathbf{L}^{-1})^\top)$  has precision  $\mathbf{L}^\top\boldsymbol{\Omega}^{-1}\mathbf{L}/\sigma^2$  and mean  $\mathbf{L}^{-1}\mathbf{r}$ , and the posterior mean of Theorem 7.6.2 is the same expression. With  $\mathbf{L} = \mathbf{v}_p^\top$  and  $\boldsymbol{\Omega} = \omega^2$ , the matrix becomes  $\mathbf{X}^\top\mathbf{X} + \mathbf{v}_p\mathbf{v}_p^\top/\omega^2$ , whose smallest eigenvalue is  $\min(\lambda_p + 1/\omega^2, \lambda_{p-1})$ : exactly the effect of a new observation of length  $1/\omega$  along  $\mathbf{v}_p$  (Proposition 26.5.1(b)).

#### Exercise B2.

Let  $x_1, x_2$  be centred with  $\|\mathbf{x}_1\|^2 = S_1$ ,  $\|\mathbf{x}_2\|^2 = S_2$  and  $\mathbf{x}_1^\top\mathbf{x}_2 = S_{12}$ , and fit the intercept and the single composite  $\mathbf{s} = \mathbf{x}_1 + \mathbf{x}_2$ . Show that its coefficient has expectation  $\beta_1 + (\beta_2 - \beta_1)(S_2 + S_{12})/(S_1 + S_2 + 2S_{12})$ . When is this  $(\beta_1 + \beta_2)/2$ ?

*Solution.* The coefficient is  $\mathbf{s}^\top\mathbf{Y}/\mathbf{s}^\top\mathbf{s}$ , because  $\mathbf{s}$  is centred and so orthogonal to  $\mathbf{1}$ . Its expectation is  $\mathbf{s}^\top(\mathbf{x}_1\beta_1 + \mathbf{x}_2\beta_2)/\mathbf{s}^\top\mathbf{s}$ , with  $\mathbf{s}^\top\mathbf{x}_1 = S_1 + S_{12}$ ,  $\mathbf{s}^\top\mathbf{x}_2 = S_2 + S_{12}$  and  $\mathbf{s}^\top\mathbf{s} = S_1 + S_2 + 2S_{12}$ . Writing  $\beta_1(S_1 + S_{12}) = \beta_1(\mathbf{s}^\top\mathbf{s}) - \beta_1(S_2 + S_{12})$  gives the formula. It

is  $(\beta_1 + \beta_2)/2$  for all  $\beta$  iff  $S_2 + S_{12} = (S_1 + S_2 + 2S_{12})/2$ , that is, iff  $S_1 = S_2$ : the composite estimates the average slope when the two regressors have equal spread, whatever their correlation.

### Exercise B3.

For  $p = 2$  and  $\mathbf{A} = \text{diag}(\lambda_1, \lambda_2)$  with  $\lambda_1 > \lambda_2$ , let  $\mathbf{x} = c(\cos \theta, \sin \theta)^\top$ . Compute  $\lambda_{\min}(\mathbf{A} + \mathbf{x}\mathbf{x}^\top)$  and show directly that it is maximized at  $\theta = \pi/2$ .

*Solution.* The trace is  $T = \lambda_1 + \lambda_2 + c^2$ , independent of  $\theta$ , and the determinant is  $D = (\lambda_1 + c^2 \cos^2 \theta)(\lambda_2 + c^2 \sin^2 \theta) - c^4 \cos^2 \theta \sin^2 \theta = \lambda_1 \lambda_2 + c^2(\lambda_1 \sin^2 \theta + \lambda_2 \cos^2 \theta)$ . The smaller eigenvalue is  $(T - \sqrt{T^2 - 4D})/2$ , which increases with  $D$  for fixed  $T$ . Since  $\lambda_1 > \lambda_2$ ,  $D$  is largest when  $\sin^2 \theta = 1$ .

### 26.5.7.3 C. Going deeper

#### Exercise C1.

Proposition 26.5.1(c) says where to put *one* new case. Repeating that advice for two cases is not optimal. Let  $p = 2$ ,  $\mathbf{A} = \text{diag}(\lambda_1, \lambda_2)$  with  $\lambda_1 > \lambda_2 > 0$ , and add two rows  $\mathbf{x}$  and  $\tilde{\mathbf{x}}$ , each of length  $c$ , so that  $\mathbf{A}$  becomes  $\mathbf{A} + \mathbf{x}\mathbf{x}^\top + \tilde{\mathbf{x}}\tilde{\mathbf{x}}^\top$ .

- (a) Show that both cases along the weak direction give  $\lambda_{\min} = \min(\lambda_1, \lambda_2 + 2c^2)$ , and that one along each eigenvector gives  $\lambda_2 + c^2$ .
- (b) Show that for *every* such pair,

$$\lambda_{\min} \leq \min\left\{\lambda_2 + 2c^2, \frac{1}{2}(\lambda_1 + \lambda_2) + c^2\right\},$$

using  $\mathbf{v}_2^\top(\cdot)\mathbf{v}_2$  for the first bound and the trace for the second.

- (c) Show that the bound is attained. When  $2c^2 \leq \lambda_1 - \lambda_2$  it is attained by (a)'s first choice. When  $2c^2 > \lambda_1 - \lambda_2$  it is attained by

$$\mathbf{x} = (\sqrt{a_1/2}, \sqrt{a_2/2})^\top, \quad \tilde{\mathbf{x}} = (\sqrt{a_1/2}, -\sqrt{a_2/2})^\top, \quad a_{1,2} = c^2 \mp \frac{1}{2}(\lambda_1 - \lambda_2),$$

neither of which lies along an eigenvector.

- (d) Take  $\lambda_1 = 2$ ,  $\lambda_2 = 1$  and  $c = 3/2$ . Compute  $\lambda_{\min}$  for the three designs of (a) and (c), and compare.

*Solution.*

- (a) If  $\mathbf{x} = \tilde{\mathbf{x}} = c\mathbf{v}_2$ , the sum  $\mathbf{A} + 2c^2\mathbf{v}_2\mathbf{v}_2^\top$  is diagonal with entries  $\lambda_1$  and  $\lambda_2 + 2c^2$ . If  $\mathbf{x} = c\mathbf{v}_1$  and  $\tilde{\mathbf{x}} = c\mathbf{v}_2$ , it is diagonal with entries  $\lambda_1 + c^2$  and  $\lambda_2 + c^2$ , whose minimum is  $\lambda_2 + c^2$  because  $\lambda_1 > \lambda_2$ .
- (b) Write  $\mathbf{M} = \mathbf{x}\mathbf{x}^\top + \tilde{\mathbf{x}}\tilde{\mathbf{x}}^\top$ , which is nonnegative definite with  $\text{tr } \mathbf{M} = 2c^2$ . By Theorem 1.7.7,  $\lambda_{\min}(\mathbf{A} + \mathbf{M}) \leq \mathbf{v}_2^\top(\mathbf{A} + \mathbf{M})\mathbf{v}_2 = \lambda_2 + \mathbf{v}_2^\top \mathbf{M} \mathbf{v}_2 \leq \lambda_2 + \text{tr } \mathbf{M} = \lambda_2 + 2c^2$ , since the eigenvalues of  $\mathbf{M}$  are nonnegative and sum to  $\text{tr } \mathbf{M}$ . And the smaller of two eigenvalues is at most their average, so  $\lambda_{\min}(\mathbf{A} + \mathbf{M}) \leq \frac{1}{2} \text{tr}(\mathbf{A} + \mathbf{M}) = \frac{1}{2}(\lambda_1 + \lambda_2) + c^2$ .
- (c) For the first case the design of (a) gives exactly  $\lambda_2 + 2c^2$ , which is then the smaller

of the two bounds and is attained. For the second,  $a_1$  and  $a_2$  are nonnegative because  $2c^2 > \lambda_1 - \lambda_2$ , they sum to  $2c^2$ , and  $\mathbf{xx}^\top + \tilde{\mathbf{x}}\tilde{\mathbf{x}}^\top = \text{diag}(a_1, a_2)$ , the off-diagonal terms cancelling. Also  $\|\mathbf{x}\|^2 = \|\tilde{\mathbf{x}}\|^2 = (a_1 + a_2)/2 = c^2$ . Then  $\mathbf{A} + \mathbf{M} = \text{diag}(\lambda_1 + a_1, \lambda_2 + a_2)$ , and both entries equal  $\frac{1}{2}(\lambda_1 + \lambda_2) + c^2$ , so  $\lambda_{\min}$  attains the second bound.

- (d) Here  $c^2 = 9/4$  and  $\lambda_1 - \lambda_2 = 1 < 2c^2$ . Both cases along  $\mathbf{v}_2$  give  $\min(2, 1 + \frac{9}{2}) = 2$ , no better than doing nothing to the strong direction. One along each eigenvector gives  $1 + \frac{9}{4} = \frac{13}{4} = 3.25$ . The optimal pair, with  $a_1 = \frac{7}{4}$  and  $a_2 = \frac{11}{4}$ , gives  $\frac{3}{2} + \frac{9}{4} = \frac{15}{4} = 3.75$ .

The greedy rule fails because after the first case the weakest direction is no longer  $\mathbf{v}_2$ , and a plan made for two cases should anticipate that. Choosing several cases at once is a design problem — here the criterion  $\lambda_{\min}$ , called E-optimality — and one-at-a-time optimization solves it only when the budget is too small to level the two eigenvalues.

### Exercise C2.

Outside information helps only if it is good enough. In the setting of [Exercise \(B1\)](#) take  $\mathbf{L} = \mathbf{v}_p^\top$  and  $\mathbf{\Omega} = \omega^2$ , and suppose the source is *wrong*:  $E(r) = \gamma_p + \delta$  with  $\gamma_p = \mathbf{v}_p^\top \boldsymbol{\beta}$  and  $\delta \neq 0$ , while  $\text{Var}(r) = \sigma^2 \omega^2$  as claimed, and  $r$  is independent of  $\mathbf{Y}$ . Write  $\kappa = 1/\omega^2$ .

- (a) Show that  $\mathbf{v}_\ell^\top \tilde{\boldsymbol{\beta}} = \hat{\gamma}_\ell$  for  $\ell \neq p$ , while  $\tilde{\gamma}_p = (\lambda_p \hat{\gamma}_p + \kappa r)/(\lambda_p + \kappa)$ . Only the weak coordinate moves.
- (b) Show that  $\tilde{\gamma}_p$  has bias  $\kappa \delta / (\lambda_p + \kappa)$  and variance  $\sigma^2 / (\lambda_p + \kappa)$ , so that

$$E(\tilde{\gamma}_p - \gamma_p)^2 = \frac{\sigma^2}{\lambda_p + \kappa} + \frac{\kappa^2 \delta^2}{(\lambda_p + \kappa)^2}.$$

- (c) Deduce that the outside information lowers the mean squared error if and only if

$$\delta^2 < \sigma^2 \omega^2 + \frac{\sigma^2}{\lambda_p} = \text{Var}(r) + \text{Var}(\hat{\gamma}_p) :$$

its error must be smaller than the two standard errors combined in quadrature.

- (d) Interpret: what happens when the source is honest, in the sense that  $\delta$  is itself drawn with mean zero and variance  $\sigma^2 \omega^2$ ? What happens when it is confident and wrong, that is when  $\omega^2$  is small and  $\delta$  is not?

*Solution.*

- (a) By [Exercise \(B1\)](#),  $\tilde{\boldsymbol{\beta}} = (\mathbf{X}^\top \mathbf{X} + \kappa \mathbf{v}_p \mathbf{v}_p^\top)^{-1} (\mathbf{X}^\top \mathbf{Y} + \kappa \mathbf{v}_p r)$ . The matrix has the eigenvectors of  $\mathbf{X}^\top \mathbf{X}$ , with eigenvalues  $\lambda_\ell$  for  $\ell \neq p$  and  $\lambda_p + \kappa$ , so its inverse is  $\sum_{\ell \neq p} \lambda_\ell^{-1} \mathbf{v}_\ell \mathbf{v}_\ell^\top + (\lambda_p + \kappa)^{-1} \mathbf{v}_p \mathbf{v}_p^\top$ . Since  $\mathbf{v}_\ell^\top \mathbf{X}^\top \mathbf{Y} = \lambda_\ell \hat{\gamma}_\ell$  (Theorem 26.1.1(a)) and  $\mathbf{v}_\ell^\top \mathbf{v}_p = 0$  for  $\ell \neq p$ , the stated coordinates follow.
- (b)  $E \tilde{\gamma}_p = (\lambda_p \gamma_p + \kappa(\gamma_p + \delta))/(\lambda_p + \kappa)$ , whose difference from  $\gamma_p$  is  $\kappa \delta / (\lambda_p + \kappa)$ . Since  $r$  and  $\hat{\gamma}_p$  are independent with variances  $\sigma^2 / \kappa$  and  $\sigma^2 / \lambda_p$  (Theorem 26.1.1(a)),

$$\text{Var}(\tilde{\gamma}_p) = \frac{\lambda_p^2 \sigma^2 / \lambda_p + \kappa^2 \sigma^2 / \kappa}{(\lambda_p + \kappa)^2} = \frac{\sigma^2}{\lambda_p + \kappa}.$$

Add the squared bias.

- (c) The inequality  $E(\tilde{\gamma}_p - \gamma_p)^2 < \sigma^2/\lambda_p$  reads

$$\frac{\kappa^2 \delta^2}{(\lambda_p + \kappa)^2} < \sigma^2 \left( \frac{1}{\lambda_p} - \frac{1}{\lambda_p + \kappa} \right) = \frac{\sigma^2 \kappa}{\lambda_p (\lambda_p + \kappa)}.$$

Multiply by  $(\lambda_p + \kappa)^2/\kappa$  and divide by  $\kappa$ :  $\delta^2 < \sigma^2(\lambda_p + \kappa)/(\kappa\lambda_p) = \sigma^2/\kappa + \sigma^2/\lambda_p$ , which is the stated bound because  $\sigma^2/\kappa = \sigma^2\omega^2 = \text{Var}(r)$ .

- (d) If the source is honest then  $E\delta^2 = \sigma^2\omega^2$ , which is strictly less than the bound, so on average the outside information helps, and it helps most where the data are weakest,  $\lambda_p$  small. If the source is confident and wrong —  $\omega^2$  small, so that  $\kappa$  is large and the answer along  $\mathbf{v}_p$  comes almost entirely from outside — the bound  $\sigma^2\omega^2 + \sigma^2/\lambda_p$  is itself small, and a modest  $\delta$  makes the mean squared error worse than plain least squares, which at least was unbiased. A stated prior variance is a claim about  $\delta$ , and the inequality says the claim has to be roughly right, not merely convenient.